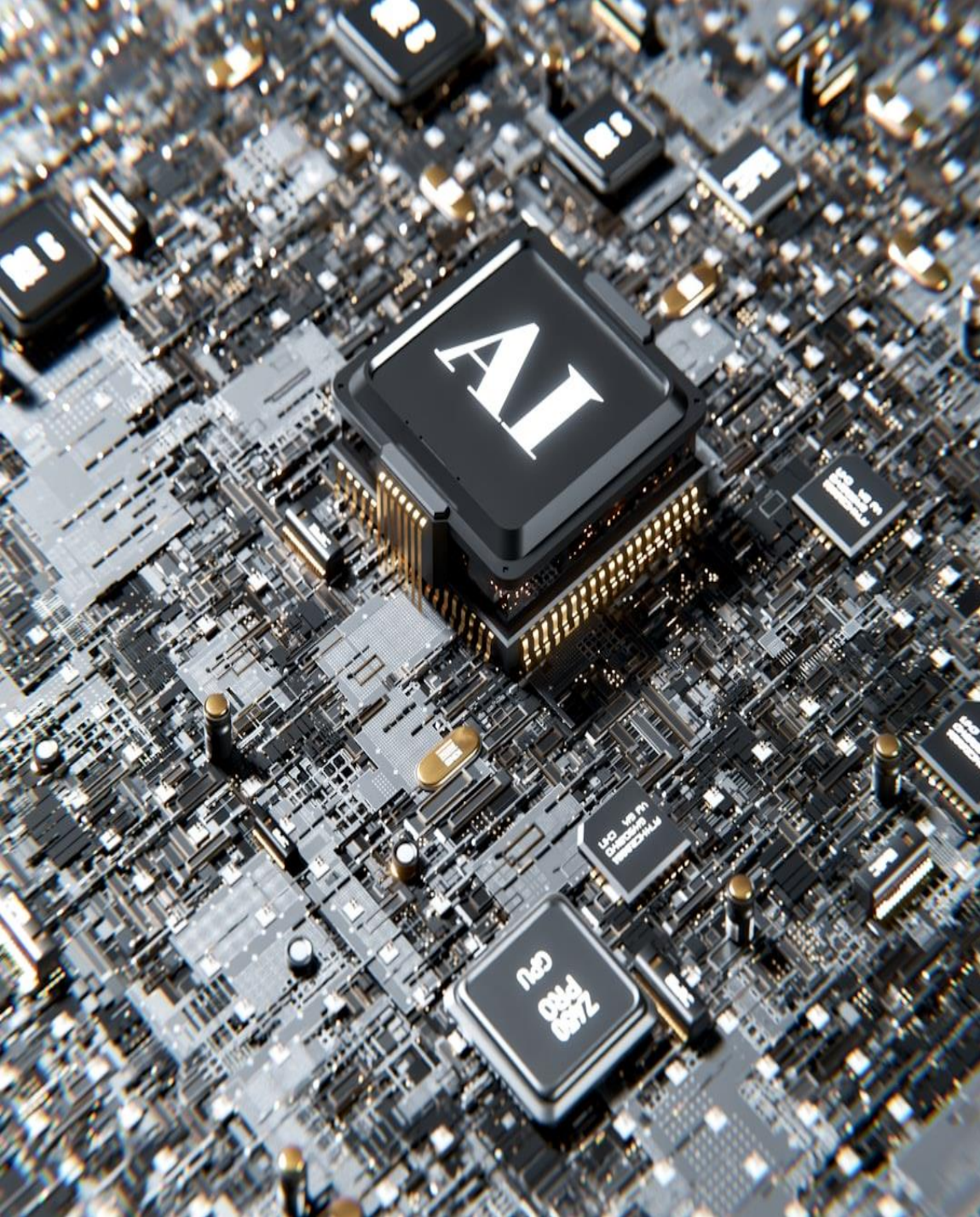




AI-Powered Resume Screening System

In a competitive job market, our AI-powered system revolutionizes resume screening. By automating the evaluation process, we enhance efficiency and reduce bias, transforming recruitment.

Challenges in Traditional Screening

High Volume of Resumes

1

Recruiters often face thousands of resumes, making manual screening time-consuming and inefficient.

Inconsistent Evaluation

2

Traditional methods can lead to subjective judgments, resulting in missed opportunities for qualified candidates.

Unconscious Bias

3

Human biases can inadvertently influence hiring decisions, affecting diversity and inclusion in the workplace.

Slow Hiring Process

4

Lengthy screening times can delay hiring, causing organizations to lose top talent to competitors.

Core Approach & Innovation

Automated Resume Parsing

Extracts essential information like skills and experience from various resume formats, streamlining the initial screening.

NLP-Based Analysis

Utilizes advanced models like BERT for deep contextual understanding, ensuring accurate matching with job descriptions.

Intelligent Resume Ranking

Generates a job match score for candidates, ranking them based on relevance to the position.



System Requirements

Developer Side	Laptop/Desktop with 8GB RAM, Python, Django, React, Database tools (e.g., MySQL Workbench)
Client Side	Any device with a modern web browser and 4GB RAM, stable internet connection, latest browser versions



Frontend Technology	Description
React.js	Interactive UI framework
Tailwind CSS	Responsive modern styling
Vite	Fast build tool
React Router DOM	Handles navigation
jsPDF / pdfjs-dist	Resume viewing and PDF generation
Firebase	Hosting and authentication
TypeScript	Typed JavaScript support

Backend Technology	Description
Flask (Python)	REST API backend framework
Flask-CORS	Cross-origin data access
PyPDF2	Resume text extraction
groq SDK	AI/NLP integration
python-dotenv	Environment configuration
Werkzeug	Flask core utilities

AI/NLP	Description
groq SDK	AI-driven resume scoring and analysis
PyPDF2	Text extraction for NLP processing

Database/Storage	Description
Firebase Firestore	Cloud NoSQL database for resumes and user data

APIs & Development Tools	Description
Flask REST API	Backend communication
Firebase API	Authentication and data handling
VS Code	Development environment
ESLint & TypeScript ESLint	Code quality tools
PostCSS & Autoprefixer	CSS optimization
Git	Version control

Deployment Strategy

Cloud Hosting

Django APIs and NLP models are hosted on platforms like AWS for high availability and performance.

Frontend Deployment

React applications are deployed on services like Netlify, ensuring fast content delivery.

Database Management

Automated backups and replication for MySQL/PostgreSQL databases enhance data resilience.

Security Measures

APIs are secured with authentication protocols, and Docker is used for consistent environments.

Ongoing Maintenance & Support

- **Data Updates**

Regular updates to training data and NLP models improve accuracy and adapt to job market trends.

- **Performance Monitoring**

Continuous monitoring of server and API performance helps identify and resolve bottlenecks.

- **Bug Fixes**

Prompt resolution of bugs and iterative development of new features based on user feedback.

- **Security Patches**

Regular application of security updates ensures data privacy and protection against vulnerabilities.

Key Features

Resume Parsing

Automated extraction of candidate information enhances efficiency in the screening process.



NLP Analysis

Deep understanding of resume content ensures accurate matching with job descriptions.



Ranking & Scoring

Objective evaluation of candidates based on relevance improves hiring decisions.



Transformative Benefits



Faster Screening

Reduces screening time by up to 80%, accelerating the hiring cycle significantly.



Bias Reduction

Minimizes human bias in initial screening, promoting fair and equitable hiring practices.



Insightful Reports

Provides data-driven insights into candidate pools, enhancing recruitment effectiveness.

Future Enhancements

CANDIDATE FEEDBACK



Integrate features for automated communication and interview scheduling with candidates.

ATS INTEGRATION



Seamless integration with Applicant Tracking Systems for a unified recruitment workflow.

RESUME CLUSTERING



Develop capabilities for bulk profiling and grouping of similar resumes for better talent management.

SENTIMENT ANALYSIS



Explore advanced NLP for deeper insights into candidate communication styles and traits.

Transforming Recruitment with AI

The AI-Powered Resume Screening System not only streamlines the recruitment process but also enhances fairness and efficiency. By leveraging advanced technologies, organizations can make informed hiring decisions, ultimately leading to a more diverse and capable workforce.

Thank
You